

Pole Placement Controller Design for an Inverted Pendulum on a Cart

1 Pole Placement Design

The design objective is to assign the closed-loop poles to specific locations in the left half of the complex plane to achieve desired transient performance.

1.1 Desired Pole Locations

In this example, the desired closed-loop poles are chosen as:

$$\text{desired_poles} = \{-2, -3, -4, -5\}.$$

- **Tuning Insight:** The chosen poles determine the system's transient response. For example, more negative poles will result in a faster response (shorter settling time), but may lead to higher control energy. Adjust these values to balance speed and control effort.

1.2 Computing the Feedback Gain

The state feedback gain K_p is computed using MATLAB's `place` function:

$$K_p = \text{place}(A, B, \text{desired_poles}).$$

This gain is used to form the closed-loop system:

$$A_{cl} = A - B K_p.$$

The resulting closed-loop eigenvalues are then verified against the desired pole locations.

2 Closed-Loop Simulation

The performance of the pole placement controller is evaluated by simulating the closed-loop response.

2.1 Initial Conditions and Simulation Setup

The simulation uses:

- Initial condition: $x(0) = [0; 0; 0.1; 0]$ (a small initial deviation in the pendulum angle).
- Simulation time vector: $t = 0 : 0.01 : 5$ seconds.

The closed-loop system is simulated using the `initial` function:

$$[y_{pp}, t, x_{pp}] = \text{initial}(sys_cl_pp, x_0, t),$$

where `sys_cl_pp` is the closed-loop state-space system constructed with A_{cl} .

2.2 Plotting Results

The simulation produces two plots:

- The first subplot shows the cart position over time.
- The second subplot shows the pendulum angle over time.

These plots help visualize the system's transient performance and stability.

3 Frequency Domain Analysis

To further assess the controller performance, frequency domain analysis is conducted for the transfer function from the control input F to the pendulum angle.

3.1 Transfer Function Extraction

The closed-loop state-space system is converted to a transfer function model:

$$\text{sys_tf} = \text{tf}(sys_cl_pp).$$

For a Single-Input Multiple-Output (SIMO) system, the transfer function from the input F to the pendulum angle (the second output in y) is extracted:

$$\text{sys_tf2} = \text{sys_tf}(2).$$

3.2 Frequency Response Plots

The following plots are generated:

- **Root Locus:** Provides insight into the movement of the closed-loop poles as the control gain varies.
- **Nyquist Plot:** Shows the frequency response and helps assess the robustness and stability margins.
- **Bode Plot:** Displays magnitude and phase information over a range of frequencies, illustrating gain and phase margins.

4 Step Response Metrics

The step response of the closed-loop system is analyzed to determine transient performance metrics such as rise time and settling time:

- The `stepinfo` function is used to compute:

Rise Time (time required to go from 10% to 90% of the final value)

Settling Time (time until the response remains within a specified tolerance)

These metrics are printed to provide quantitative performance evaluation.

5 Summary of Tuning and Performance

- **Desired Poles:** The choice of $\{-2, -3, -4, -5\}$ determines the speed and damping of the closed-loop response. More negative values yield faster response but may increase control effort.
- **Feedback Gain K_p :** Calculated via the `place` function, K_p directly affects the closed-loop dynamics.
- **Closed-Loop Eigenvalues:** Verifying that the eigenvalues of $A - BK_p$ match the desired poles ensures that the design objective is met.
- **Frequency Domain Analysis:** Root locus, Nyquist, and Bode plots provide insight into stability margins and robustness.
- **Step Response Metrics:** Rise time and settling time provide a quantitative measure of the transient performance.

Adjusting the desired pole locations is the primary means of tuning this controller. A designer must balance between a rapid response (requiring more negative poles) and control energy limitations or actuator saturation.

6 Conclusion

The pole placement controller designed herein uses state-feedback to assign closed-loop poles at predetermined locations. By carefully selecting desired poles and computing the corresponding gain using the `place` function, the closed-loop system is ensured to have the desired transient response. Simulation results and frequency-domain analyses verify the performance, and step response metrics quantitatively evaluate system behavior. Fine-tuning the desired poles is critical to achieve the optimal balance between response speed, damping, and control effort.